
An Automated Sea Urchin Counting Pipeline for Kelp Forest Restoration Monitoring

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Abstract

Kelp forests along the northern California coast have undergone a large-scale transition to urchin barrens, and targeted removal of purple sea urchins has become a primary focus of restoration efforts. Prioritizing removal effort requires repeatable estimates of urchin density, but manual counts from diver-collected transect video are slow, costly, and difficult to standardize across surveys. We developed an open, inspectable pipeline that counts individual sea urchins from underwater transect video by separating frame-level detection from multi-object tracking. A pre-trained R-CNN detector produces per-frame bounding boxes; a tracking stage then links detections across frames using a constant-velocity Kalman filter and Hungarian matching assignment, with track confirmation and aging rules to suppress spurious short-lived tracks. We ran the pipeline on 17 transect videos spanning three survey sites and two capture formats, seven of which had manual reference counts suitable for quantitative validation. Across those seven videos the pipeline achieved a mean absolute error of 81.6 urchins, a mean absolute percentage error of 14.2%, and a mean bias of +63 urchins. Tracker settings were selected per video, so these figures bound from above what a single fixed configuration would achieve. Human counters disagreed with their own consensus by 9.5%, the pipeline by 7.5%. On out-of-distribution footage recorded at a different resolution, aspect ratio, and frame rate, the default configuration undercounted by 62.4%; parameter rescaling and the distance-based fallback reduced this to 8.7% without retraining the detector. We release the source code, parameter profiles, and a diver filming protocol intended to reduce format-driven variability in future surveys.

1 Introduction

Kelp forests are among the most productive and structurally important ecosystems in the ocean. Formed by large canopy-forming species such as bull kelp (*Nereocystis luetkeana*), these forests create underwater habitats that support more than a thousand marine species, including fish, invertebrates, marine mammals, and seabirds, while also delivering tangible benefits to nearby human communities. Healthy kelp forests reduce wave energy, help protect coastlines from erosion, and locally buffer ocean acidification. Despite this importance, kelp forests are sensitive to changes in water temperature, nutrient availability, and invertebrate grazing pressure.

Worldwide, this vulnerability has translated into substantial losses. Estimates suggest that forty to sixty percent of kelp forests have disappeared in the last fifty years, with some regions experiencing near-total collapse (McPherson et al., 2021). These losses are not simply short-term fluctuations: in many places kelp has failed to recover even after environmental conditions improved.

Along the northern California coast this pattern of collapse and non-recovery is especially clear. [McPherson et al. \(2021\)](#) document a rapid, coast-wide decline in bull kelp beginning around 2014, coinciding with an intense and prolonged marine heatwave in which reduced nutrient availability weakened kelp growth and reproduction. Warming alone does not explain the scale or persistence of the loss; the authors describe a large-scale shift in the structure of a kelp forest ecosystem, emphasizing that kelp forests transitioned into a fundamentally different state rather than temporarily declining. During the same period, populations of sunflower sea stars (*Pycnopodia helianthoides*), a key predator of purple sea urchins (*Strongylocentrotus purpuratus*), fell dramatically due to a wasting disease ([McPherson et al., 2021](#)). With predation pressure gone, purple sea urchin populations expanded and began grazing kelp at rates far exceeding its ability to recover, and kelp forests transformed into urchin barrens, a self-reinforcing state. Even when temperatures cooled and nutrient conditions improved, kelp often failed to return, because dense urchin populations consumed young kelp before it could reach sexual maturity and reproduce.

The persistence of urchin barrens carries ecological and economic consequences. Kelp forests create vertical structure supporting diverse fish assemblages, particularly juvenile fish that rely on kelp for shelter during vulnerable life stages; as kelp disappears, dependent species decline, leading to simplified ecosystems with lower biodiversity and reduced productivity. The decline of kelp has contributed to collapses in red sea urchin fisheries, displaced by purple sea urchin, and has left other species increasingly vulnerable. Red abalone (*Haliotis rufescens*) in particular have shown lasting sublethal effects: surviving individuals exhibited body condition declines and a 99% drop in mature egg production, demonstrating how the loss of a foundation species can suppress population recovery well beyond the initial die-off ([Rogers-Bennett et al., 2021](#)).

Restoration has struggled to keep pace with the scale of loss. Efforts are often small, expensive, and labor intensive, and most kelp restoration projects are less than one hectare in size and cost hundreds of thousands of dollars per hectare ([Eger et al., 2022](#)). Over the last decade, numerous studies have demonstrated that targeted removal of sea urchins is the most effective and widely implemented strategy for restoring kelp forests, with success reported in regions such as California and Australia ([Williams et al., 2021](#); [Ling et al., 2009](#)). Across these efforts, removal has relied primarily on subtidal hand-harvesting or culling conducted by divers ([Ward et al., 2022](#); [Eger et al., 2022](#); [Miller & Shears, 2023](#); [McHugh et al., 2025](#)). Since 2020, restoration along the North Coast has focused on reducing urchin population density and lessening the impact of urchins overgrazing recovering kelp ([Ward et al., 2022](#); [McHugh et al., 2025](#)). However, this region is remote, rugged, and an exposed open-coast environment characterized by frequent wind and wave events ([Ward et al., 2022](#)), which limits the number of days on which safe diving can be conducted and emphasizes the need for efficient ways of quantifying urchin removal effort.

Monitoring at this scale depends on turning diver-collected video into defensible density estimates. Existing practice relied on a closed-source vendor executable whose internal parameters could not be inspected, which makes it difficult to tell whether a change in counts between surveys reflects ecological change or a change in the tool.

Contributions.

- We rebuild the counting workflow as an open system that separates frame-level detection from multi-object tracking, exposing every parameter and intermediate output.
- We quantify agreement with manual reference counts and against the vendor executable, and establish an inter-counter reliability baseline showing that the pipeline sits inside the range of human disagreement.
- We diagnose how accuracy responds to footage recorded in formats the detector was not tuned for, and show that the loss is recoverable at the tracking stage without retraining the detector.

2 Methods

2.1 Data and workflow motivation

The primary data are underwater transect videos collected by The Nature Conservancy along the northern California coast. Each video is processed frame by frame so that an object detector can be applied to individual images. For each frame, the detector produces bounding boxes, confidence scores, and class labels for visible urchins. These frame-level detections are stored in JSON format and passed to a pathway script for filtering, tracking, and counting.

We began by examining the existing vendor executable and the outputs it generated on a set of test videos. This inspection showed that the executable produced frame-level urchin detections, but that detection output alone was not sufficient to generate a final aggregated count: the counting step required a separate tracking procedure to link detections across frames and identify which detections corresponded to the same individual urchin.

We therefore reimplemented the workflow as two scripts. The first runs the detection model and writes frame-level results. The second processes those detections, tracks bounding boxes across frames, and produces a final count of unique sea urchins. Because detection and aggregation are separated, each stage can be inspected, debugged, and tuned independently for videos with different resolutions, frame rates, and visual conditions. Figure 1 summarizes the end-to-end workflow.

2.2 Automated processing infrastructure

The pipeline runs without human intervention once footage is uploaded (Figure 2). A diver returns from a transect dive with GoPro footage and, lacking direct cloud access, uploads it to a shared Box folder. The upload triggers a transfer into an Amazon S3 bucket, which fires two events in parallel: an EventBridge notification that starts, or confirms the status of, a dedicated EC2 processing instance, and a message placed on an SQS queue recording the new file’s S3 path. A lightweight Lambda function validates that the uploaded object is an MP4 before starting the instance, and leaves it alone if it is already running, so that idle compute is not billed between dive days.

Once the instance is live, a worker process polls the queue as a to-do list, pulling messages in batches, downloading the referenced files, and discarding duplicate or transient notifications using retry logic with exponential backoff. This mattered more than expected in practice: intermediate renaming and re-uploading of files during transfer produced spurious notifications for files that briefly appeared missing, and without the retry logic these would have been dropped. Outputs are written to three places: a CSV of per-video counts, density estimates, and metadata; the annotated video itself, with tracked bounding boxes burned in for visual quality assurance; and a running master file across every video processed to date.

2.3 Detection

For every detected urchin the model outputs a bounding box, a confidence score, and a class label, stored in JSON and associated with the corresponding frame number. Two-stage detectors such as Faster R-CNN perform inference on individual frames; they identify where objects appear, but do not by themselves determine whether detections across multiple frames belong to the same individual (Girshick, 2015; Ren et al., 2015; Redmon & Farhadi, 2018). The detector is treated throughout as a fixed component; we did not retrain it.

2.4 Multi-object tracking

The pathway script follows a tracking-by-detection approach, a common framework when frame-level detections are already available. The detector identifies where objects appear in each frame, while the tracker links detections across consecutive frames and decides which belong to the same individual. This structure is the basis of SORT-family trackers, including SORT and DeepSORT (Bewley et al., 2016; Wojke et al., 2017).

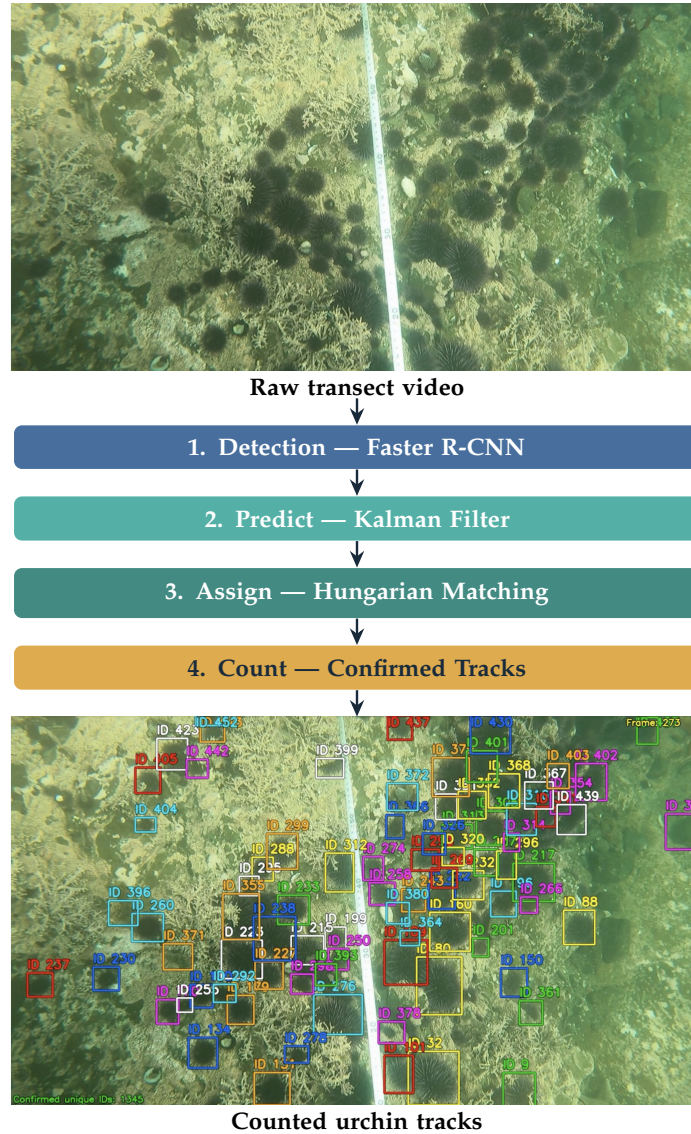


Figure 1: End-to-end counting workflow. A raw transect frame (top) passes through Faster R-CNN detection, Kalman filtering, Hungarian multi-object tracking, and a counting step to produce an annotated transect frame (bottom). Both frames show the same scene, before and after tracking; each colored box in the lower panel is one confirmed unique urchin. Note that identity, and therefore the count, is established at the tracking stage rather than by the detector: a single urchin is detected in many consecutive frames, and only a track confirmed across those frames becomes one counted individual. This is why changes confined to the tracking stage recover so much of the count without altering the detector.

The tracker maintains a set of active tracks. For each new frame it predicts the next position of every active track using a motion model, compares those predictions with new detections, builds an assignment cost matrix, and uses the Hungarian algorithm to find the best matching. Matched tracks are updated, unmatched tracks are aged, and unmatched detections can start new tracks.

Before tracking begins, detections with confidence scores below a chosen `score_threshold` are removed, and the remainder are filtered with non-maximum suppression based on intersection-over-union (IoU). Confidence thresholding and non-maximum suppression

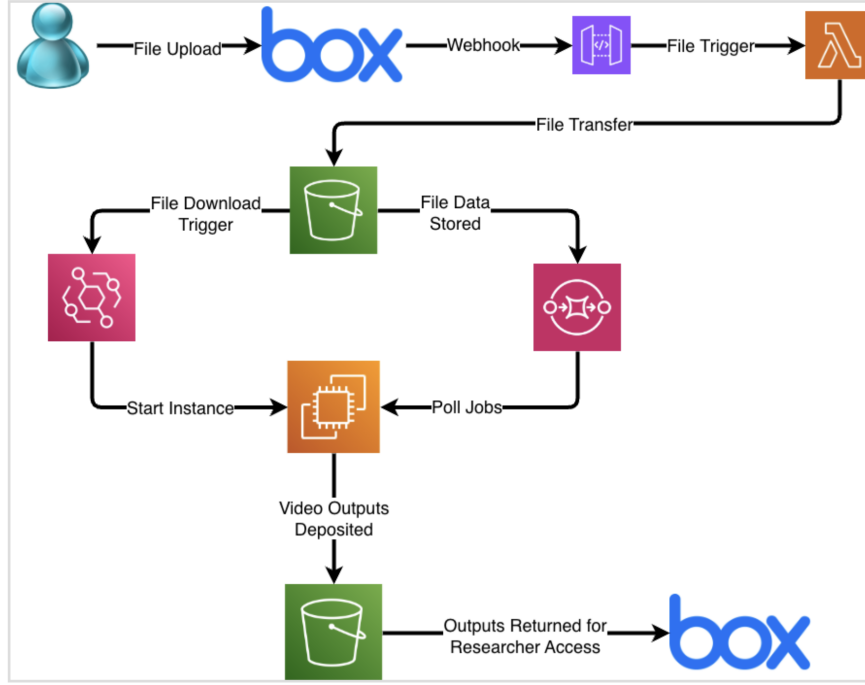


Figure 2: Architecture of the automated processing pipeline. A diver upload triggers a transfer into cloud object storage, which starts a processing instance and enqueues the new file; a worker then downloads the video, runs detection and tracking, and writes counts, density estimates, and an annotated review video back to storage. Note that the two branches fired by the upload are independent: instance startup and job queueing proceed in parallel, so a batch of videos arriving together is processed without repeated instance starts.

reduce false positives and duplicate detections before downstream analysis (Redmon et al., 2016; Zhou et al., 2024).

Detections are then linked across frames by comparing the predicted bounding box of each active track with each new detection using IoU. These values are converted into a cost matrix in which lower cost indicates a better match, and the Hungarian algorithm finds the assignment minimizing total cost (Kuhn, 1955). A matched pair is accepted only if its IoU exceeds a configurable `match_iou` threshold. This combination of Kalman prediction (Kalman, 1960), IoU-based costs, and Hungarian assignment follows the structure of SORT-style tracking (Bewley et al., 2016).

Raw tracking can produce more tracks than the true number of urchins, because short-lived false positives or unstable detections may briefly appear as separate tracks. A track is therefore counted only if it accumulates at least `min_hits` matched detections, and tracks that remain unmatched for more than `max_age` consecutive frames are removed from the active set. Lower `min_hits` admits more false positives, while higher `min_hits` may exclude real urchins that appear only briefly; higher `max_age` can bridge short detector misses, while lower `max_age` reduces the chance of keeping stale tracks alive.

2.5 Distance-based association fallback

IoU-based matching works well when detections of the same urchin overlap across consecutive frames. This assumption fails when the camera moves quickly, when the video has a lower frame rate, or when the detector produces small shifts in bounding-box position. In these cases two detections may belong to the same urchin but have little or no overlap;

because IoU becomes zero, the tracker treats them as unrelated objects, and a single urchin pathway breaks into multiple short fragments.

To reduce this fragmentation we added a distance-based fallback to the assignment step. The tracker still prioritizes IoU-based matches when bounding boxes overlap sufficiently. When IoU is too low, the fallback allows a match if the predicted track center and the detected box center fall within a configurable distance threshold (`max_step`), giving the tracker a second way to preserve continuity when overlap-based matching fails.

2.6 Parameter selection

The detector and tracker interact in non-obvious ways: a stricter confidence threshold reduces noise but risks losing faint urchins, while a looser matching IoU reduces fragmentation but risks merging distinct individuals. We therefore selected settings using a 256-combination grid search across score threshold, non-maximum suppression IoU, matching IoU, `max_age`, and `min_hits`.

Settings were chosen separately for each video rather than fixed within a site or a capture format. Table 1 reports the configuration used for every validated video; no two in-distribution videos share a score threshold, and the four Big River videos differ in three of the five parameters. Two consequences follow, and we state them before reporting any accuracy. First, the errors we report are those achievable when each video is tuned individually, so they are an upper bound on what a single fixed configuration would deliver in routine use. Second, the grid search was run over the same seven videos used for evaluation, so the reported errors are optimistically biased by an unknown amount; with seven videos we could not hold out an independent set.

2.7 Counting, density, and validation

After tracking, confirmed tracks are counted as individual sea urchins. These counts are converted to densities using the surveyed area, which is what restoration crews need for prioritizing removal sites, since a raw count means little without knowing how much reef it came from.

Surveyed area is the product of transect length and the width of the strip the camera sees. Rather than assume a nominal width, we measured it from the transect tape, a known-width object visible in most frames: its apparent width in pixels gives the ground scale, and frame width in meters follows from the frame width in pixels. Each transect's density is therefore computed from its own measured width, and the reported interval reflects the variation between frames within that transect.

Manual reference counts were produced by four of the authors, each of whom counted a subset of the videos independently by reviewing the footage frame by frame. The reference count for a video is the mean of the counters who scored it; four videos were scored by all four counters, and three by two counters. We compared automated counts against these reference counts using mean absolute error (MAE), mean absolute percentage error (MAPE), mean bias, and Bland–Altman limits of agreement (Bland & Altman, 1986).

Because the reference is itself a human measurement, we also quantified how much the counters disagreed with one another, using the four videos scored by all four. Three complementary summaries are reported: a leave-one-out comparison, in which each counter is scored against the mean of the other three and which is the direct analogue of the pipeline being scored against a consensus it does not contribute to; the absolute percentage difference between every pair of counters; and the intraclass correlation coefficient for absolute agreement, $ICC(2,1)$ for a single counter and $ICC(2,k)$ for the mean of four.

2.8 Functional density along the transect

A single density per transect collapses a whole swim into one number, and that number is identical whether the urchins were spread evenly over the reef or piled into a single patch. Removal crews act on where the animals are, not only on how many there were, so we also

estimate density as a function of position along the transect. We call this the functional urchin density and write it $d(s)$, with s the distance in meters from the start of the transect.

Both ingredients are measured with error. Per-frame counts fluctuate as animals are occluded by kelp or pass through the frame edge, and the per-frame ground area fluctuates as the camera rises and falls above the substrate. A natural estimate is therefore the ratio of the two, each smoothed in position,

$$d(s) = \frac{\hat{c}(s)}{\hat{a}(s)}, \quad (1)$$

where $\hat{c}$ and $\hat{a}$ are Nadaraya–Watson kernel smoothers (Nadaraya, 1964; Watson, 1964) of the per-frame detection count and the per-frame ground area. In this application, Gaussian kernels were used and bandwidths are set as a fraction of transect length, 3% for counts and 4% for area, with floors of 0.18 m and 0.25 m; area gets the wider bandwidth because camera altitude drifts more slowly than the animals are encountered. The ratio is evaluated on a 240-point grid spanning the transect.

Smoothing the numerator and denominator separately, instead of smoothing the per-frame ratio directly, is a deliberate choice. Ground area is only measurable on frames where the tape is legible enough to give a scale, which is between 39 and 105 frames per video, whereas counts are available on every frame. Forming the ratio first would discard every frame without a usable tape measurement, throwing away the great majority of the count information. Smoothing separately lets each quantity use all the frames on which it is actually observed.

Note we assign position s by frame index scaled to the transect length, which assumes the diver advanced at a constant pace. Curves were obtained for 15 of the 17 videos. The two exceptions were filmed with the camera rotated into portrait orientation, and in them the tape lies across the frame instead of running down.

3 Results

We ran the pipeline on 17 transect videos in total, covering North Coast 2025 surveys, Big River 2026 pre-removal blocks, and Noyo 2026 post-intervention surveys. These included in-distribution footage recorded at 1920×1080 resolution, 16:9 aspect ratio, and 60 fps, and out-of-distribution Big River footage recorded at 5312×4648 resolution, 8:7 aspect ratio, and 30 fps. Of the 17, seven had manual reference counts suitable for quantitative validation (Table 1).

3.1 Validation against manual counts

Across the seven validated videos the pipeline achieved a MAE of 81.6 urchins, a MAPE of 14.2%, and a mean bias of +63 urchins. Bland–Altman limits of agreement were approximately -174 to $+300$ urchins (Figure 4). These figures are per-video tuned, in the sense set out above: each video was counted using the settings in Table 1, selected for that video, so they place an upper bound on what a single fixed configuration would achieve. Because the validation set is small, the limits of agreement should also be read as provisional.

The most accurate result was Big River Block-5 Heading-0, where the pipeline counted 213 urchins against a manual reference count of 214, an error of -0.5% . The densest video, 06-17-25 Transect 1, was the largest absolute miss: 1,345 pipeline-counted urchins against a manual reference count of 1,033, an overcount of $+30.2\%$ (Figure 3). That reference is also the least reliable of the seven, resting on two counters who differed by 43% of their mean.

Overall the pipeline tended to overcount rather than undercount. This positive bias reflects a deliberate tuning choice: for this monitoring application undercounting is more damaging than overcounting, because an underestimated density could make a restoration site appear less urgent than it is.

Table 1: Per-video validation results with the tracker settings used for each. Parameters were selected per video rather than fixed within a site or capture format, so Δ is the accuracy achievable when each video is tuned individually, not the accuracy of a single deployable configuration.

Video	Tracker settings					Counts		
	Score thr.	Min hits	Max age	Max step	Dist. fb	Manual	Pipeline	Δ (%)
06-16-25 Transect 1	0.30	50	6	80	no	309	362	+17.2
06-17-25 Transect 1	0.25	50	6	80	no	1,033	1,345	+30.2
07-10-25 Transect 5	0.20	50	6	80	no	334	339	+1.5
BR Block-5 Heading-0	0.20	25	30	160	yes	214	213	-0.5
BR Block-5 Heading-90	0.20	25	20	160	yes	352	391	+11.1
BR Block-5 Heading-180	0.15	15	20	160	yes	735	671	-8.7
BR Block-6 Heading-180	0.15	25	20	160	yes	318	415	+30.5

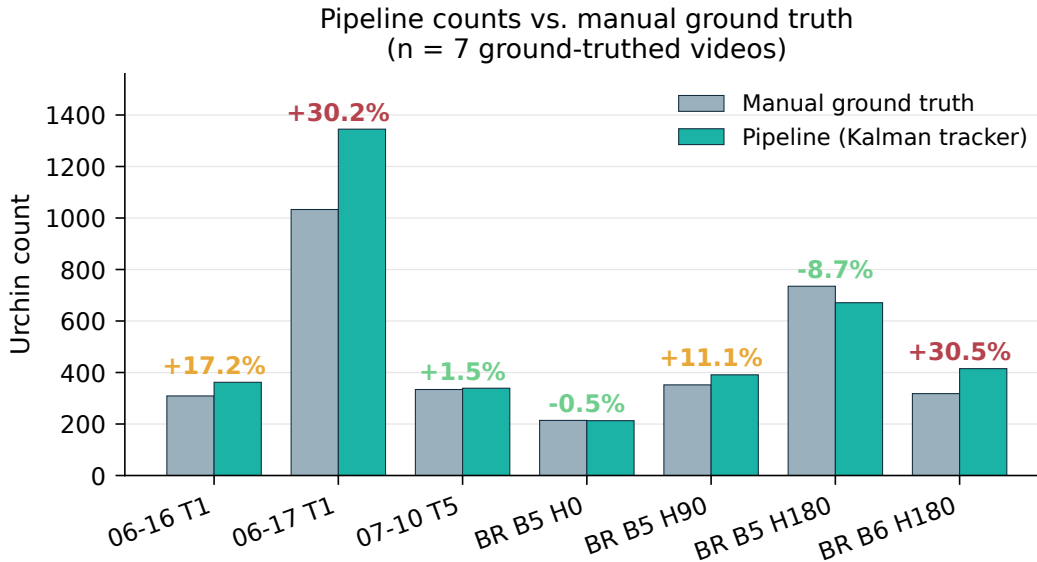


Figure 3: Pipeline counts compared with manual ground truth for the seven videos with multi-counter manual counts. Annotated percentages give the per-video signed error. Note that error grows with density rather than staying proportionally constant: the densest video, at a reference count of 1,033 urchins, carries both the largest absolute error (+312) and, with Big River Block-6 Heading-180, essentially the largest relative one. Five of the seven errors are positive, which is the deliberate bias toward overcounting described in the text.

3.2 Agreement among human counters

The reference counts are themselves human measurements, and they varied substantially. Across the four videos scored by all four counters (Table 2), the coefficient of variation among counters ranged from 3.3% on 06-16-25 Transect 1 to 15.7% on Big River Block-5 Heading-0, where individual counts spanned 181 to 250 urchins on the same footage. The median absolute difference between any two counters was 8.4%, reaching 32.0% at worst.

Scored against the mean of the other three, a single counter departed from consensus by 26.6 urchins on average, a MAPE of 9.5%, with limits of agreement of -66 to +66 urchins. There was no systematic direction to this disagreement across counters as a group (mean bias +0.0), though individual counters differed: the most conservative ran 4.7% below the others on average and the most liberal 5.0% above. Agreement was moderate for a single counter, $ICC(2,1) = 0.816$, and high for the four-counter mean, $ICC(2,k) = 0.947$.

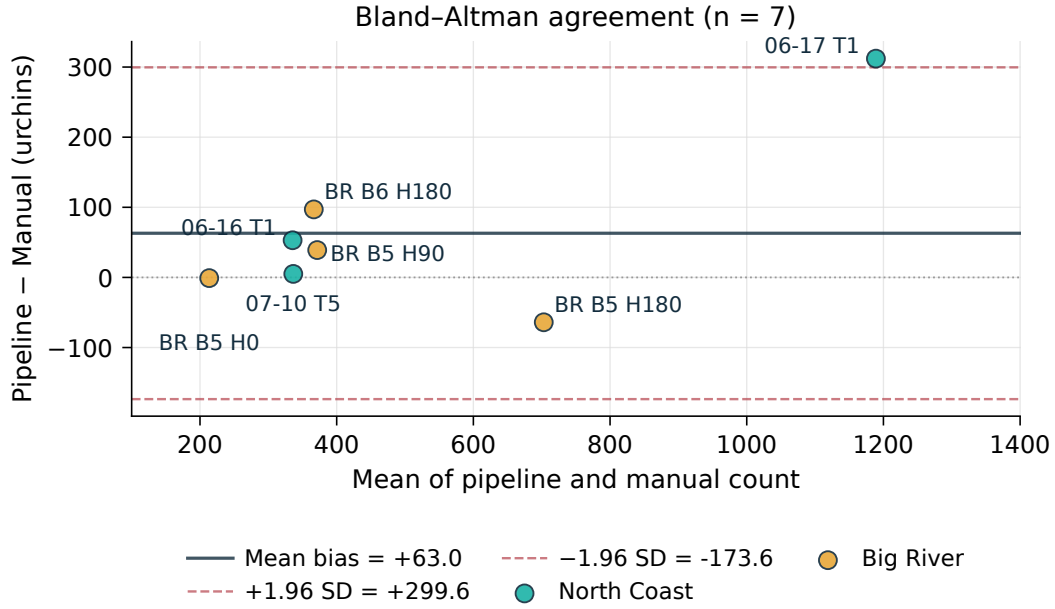


Figure 4: Bland–Altman agreement between pipeline and manual ground truth across the seven ground-truthed videos. The solid line marks the mean bias (+63.0 urchins); dashed lines mark the 1.96 SD limits of agreement. Note that the differences scale with the mean count ($r = +0.71$), so the tracker shows proportional bias: it overcounts denser transects by more, in both absolute and relative terms. The single largest deviation (+312 urchins) is the densest video, whose reference count rests on only two counters who differed by 43% of their mean, so part of that gap may sit in the reference rather than in the tracker.

Table 2: Independent counts by the four human counters on the four videos scored by all of them. Counters are anonymised and ordered consistently across rows. CV is the coefficient of variation among the four counts. Note that disagreement is not uniform: it is smallest on the sparsest North Coast transect and largest on Big River Block-5 Heading-0, where the same footage yielded counts from 181 to 250.

Video	Counter				Mean	SD	CV (%)
	1	2	3	4			
06-16-25 Transect 1	299	313	302	321	308.8	10.1	3.3
07-10-25 Transect 5	322	357	313	345	334.2	20.3	6.1
BR Block-5 Heading-0	181	190	250	235	214.0	33.7	15.7
BR Block-5 Heading-90	405	310	350	344	352.2	39.3	11.2

On these same four videos the pipeline achieved a MAE of 24.5 urchins and a MAPE of 7.5%, against 26.6 and 9.5% for a human counter. The pipeline sits inside the range of human disagreement rather than outside it, although it carries a positive bias (+24.0 urchins) that the counters as a group do not.

3.3 Estimated urchin densities

Converting counts to densities places the surveys on a comparable footing (Table 3). Taking the point estimate from each transect’s median frame width, the six Big River pre-removal transects average 30.4 urchins m^{-2} , the three North Coast transects 35.7, and the three

Table 3: Estimated urchin densities for the twelve videos with both a recorded transect length and a measured frame width. Counts are pipeline counts. Density is the point estimate, count divided by length times the median measured frame width. The range is not measurement error in that median: frame width changes along a swim as the camera rises and falls, by 12% to 81% of the median within a single transect, and the range gives the density implied by the lower and upper quartiles of the per-frame widths. Note that the Noyo post-intervention transects are several times sparser than the Big River pre-removal ones, a gap far larger than that within-transect variation.

Site	Video	Length (m)	Count	Frame width (m)	Density (m ⁻²)	Range (m ⁻²)
North Coast	06-16-25 Transect 1	30	362	0.94	12.8	10.9–12.8
	06-17-25 Transect 1	30	1,345	0.81	55.3	42.3–55.3
	07-10-25 Transect 5	10	339	0.87	39.0	36.1–58.4
Big River (pre-removal)	BR Block-5 Heading-0	10	213	1.24	17.2	13.9–19.0
	BR Block-5 Heading-90	10	391	1.12	34.9	29.6–43.9
	BR Block-5 Heading-180	10	671	1.44	46.6	43.9–49.3
	BR Block-5 Heading-270	10	260	1.36	19.1	17.1–19.3
	BR Block-6 Heading-0	10	495	1.22	40.6	37.8–44.6
	BR Block-6 Heading-180	10	415	1.74	23.9	23.9–27.1
Noyo (post-removal)	Noyo Block-8 Heading-0	10	71	0.96	7.4	6.3–7.6
	Noyo Block-8 Heading-180	10	39	0.81	4.8	4.1–5.1
	Noyo Block-9 Heading-0	10	110	0.87	12.6	12.4–18.6

Noyo post-intervention transects 8.3. Because frame width varies along each swim, the corresponding quartile ranges are 27.7–33.9, 29.7–42.2 and 7.6–10.4.

The Big River and Noyo contrast is large relative to the measurement uncertainty. Pairing the most conservative Big River bound with the most liberal Noyo bound still leaves Big River more than twice as dense. Big River and Noyo are, however, different sites surveyed on different dates, and the Noyo transects were filmed after removal while the Big River transects were filmed before it, so the comparison is observational and site and treatment cannot be separated with these data. Densities could not be computed for the five videos whose transect lengths were not recorded.

3.4 Density structure along and between transects

The functional densities of Section 2.8 recover the scalar estimates while adding the structure those estimates average away. Across the twelve transects for which both are available, the mean of $d(s)$ and the scalar density correlate at $r = 0.910$, with a median ratio of 0.98 and a median absolute difference of 2.8 urchins m⁻². The two constructions share no machinery beyond the detector: one divides a tracked count by a median frame width, the other integrates a smoothed ratio of per-frame counts to per-frame areas. Their agreement therefore corroborates both.

What the scalar density does not capture is how uneven the urchin distribution is (Figure 5). Within a single transect, $d(s)$ has a median coefficient of variation of 0.54, and the median ratio of its highest to its lowest value is 7.9. A crew swimming a 10 m line is therefore passing through stretches that vary considerably in density, and the peak is not reliably found at the start of a transect: across the 15 curves it falls anywhere between 5% and 100% of the way along.

The unevenness is not anchored to any particular part of the swim. Rescaling each curve by its own mean and averaging across the fifteen leaves a profile that is flat within noise, and neither the first nor the last 15% of a transect differs reliably from its middle. The patches are present but their position is not predictable. A shorter transect would therefore be no more biased than a longer one, though it would average over less ground and return a noisier estimate. Extra length helps less than one might expect, because neighboring stretches of

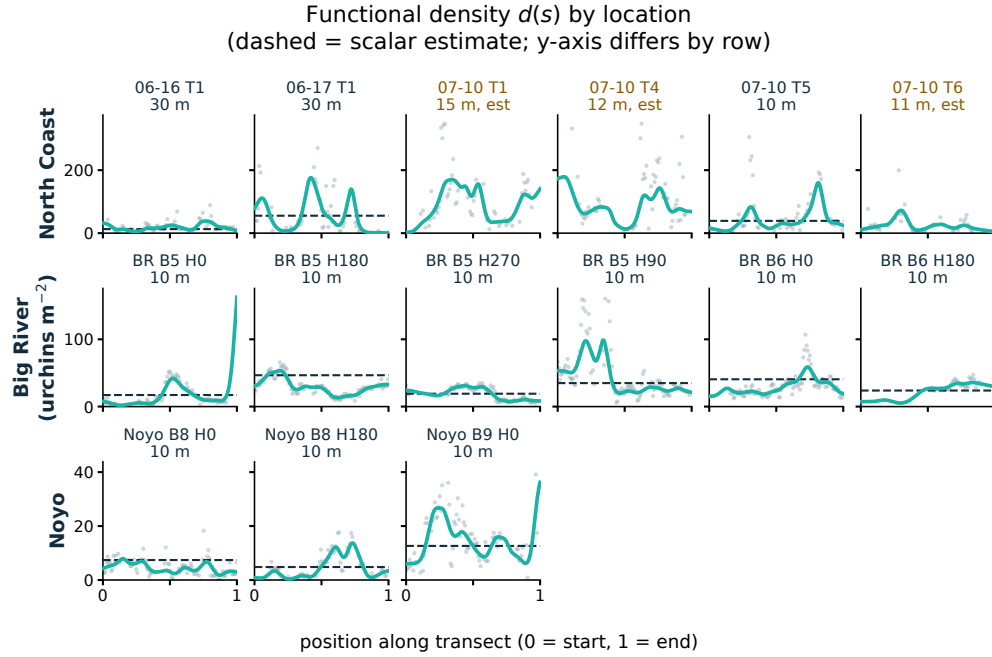


Figure 5: Functional density $d(s)$ for the 15 transects with trackable tape, grouped by survey location. Gray points are the per-frame estimates the curve is smoothed from; the dashed line is the scalar density of Table 3, where a recorded length makes one available. Titles in amber mark transects whose length is estimated from video duration rather than recorded. The y-axis is shared within a row but not between rows, because Noyo’s entire range falls inside the lowest tenth of North Coast’s. Note that almost every curve crosses its own dashed line several times: the scalar density is a fair average but describes no particularly consistent stretch of the transect.

reef tend to look alike and each additional meter adds less new information than the last: halving the distance swum worsens the density estimate only from about 25% to 31%. The design implication is that survey effort is better spent on recording additional transects than on longer ones.

3.5 Comparison with the vendor executable

We compared the pipeline with the vendor-supplied executable on the four videos for which manual reference counts, pipeline counts, and executable counts were all available (Table 4). Because executable outputs were not available in directly comparable form for every video, this is a limited baseline comparison rather than a full 17-video evaluation.

Across these four videos the executable showed a much wider spread of errors. Its error ranged from -54.2% on 07-10-25 Transect 5 to $+76.1\%$ on Big River Block-5 Heading-180, whereas our pipeline ranged from -8.7% to $+30.2\%$ on the same videos. On 06-17-25 Transect 1, however, the executable was the more accurate of the two, counting 1,034 against a manual reference count of 1,033, while our pipeline counted 1,345. Across these four videos the executable therefore spans roughly 130 percentage points of error against our 39, but it is not uniformly worse.

3.6 Effect of capture format and parameter tuning

The most challenging videos were the Big River videos, which differed from the in-distribution footage in resolution, aspect ratio, and frame rate. The default in-distribution tracking profile did not transfer to this footage.

Table 4: Head-to-head comparison against the vendor-supplied executable, across the four videos for which manual ground truth, pipeline output, and executable output are all available.

Video	Manual ground truth	Pipeline (count)	Pipeline error	Executable (count)	Executable error
06-16-25 Transect 1	309	362	+17.2%	386	+24.9%
06-17-25 Transect 1	1,033	1,345	+30.2%	1,034	+0.1%
07-10-25 Transect 5	334	339	+1.5%	153	-54.2%
BR B5 Heading-180	735	671	-8.7%	1,294	+76.1%

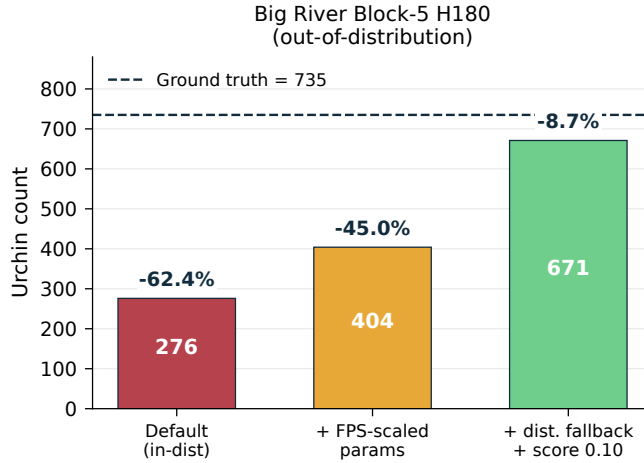


Figure 6: Progressive recovery of counts on Big River Block-5 Heading-180, recorded outside the format the detector was tuned for. Default parameters undercount the manual reference count of 735 by 62.4%. Rescaling tracking parameters for frame rate reduces the gap to -45.0%. Adding the distance-based association fallback and lowering the score threshold to 0.10 reduces the remaining gap to -8.7%. Note that the final step recovers more than twice as many urchins as the frame-rate rescaling (267 against 128 of the 459 initially missed), but it changes two settings at once, so the distance fallback’s isolated contribution cannot be read from this figure.

The failure mode was track fragmentation rather than complete detector failure. Annotated review videos showed the detector still producing many candidate urchin boxes, but the tracker often split the same urchin into several short tracks. Because tracks must survive long enough to pass the `min_hits` confirmation rule, many fragments were discarded before final counting, producing severe undercounts.

Big River Block-5 Heading-180 illustrates this most clearly (Figure 6). Under the default profile the pipeline counted only 276 urchins against a manual count of 735, an error of -62.4%. Rescaling tracking parameters for frame rate alone improved the count to 404, reducing the error to -45.0%. The fully tuned profile, combining rescaled parameters, a lower score threshold, and the distance-based fallback, increased the count to 671, reducing the error to -8.7%. The detector was unchanged throughout.

Comparable tuning was then carried out for the remaining Big River videos, with settings chosen individually for each. This substantially improved several headings, including Block-5 Heading-0 (-0.5%), Block-5 Heading-90 (+11.1%), and Block-5 Heading-180 (-8.7%), although Block-6 Heading-180 still overcounted by +30.5% at settings differing from Block-5 Heading-180 only in `min_hits`. Per-video tuning therefore removed the major format-driven failure without eliminating all video-specific error, and the residual spread indicates that a single Big River configuration would not have reproduced these results.

4 Discussion

A central contribution of this work is an open, inspectable, reproducible tool for counting sea urchins from underwater transect video. Compared with the vendor executable it replaces, the pipeline exposes every major step: detection outputs, pathway generation, tracker parameters, intermediate files, and annotated review videos. This transparency matters for restoration monitoring specifically, because count differences between surveys should reflect real environmental change rather than hidden changes in the counting tool. A monitoring program that cannot distinguish the two cannot defend its own trend estimates.

Setting that accuracy against human performance is what makes it interpretable. Four counters working from the same footage disagreed with their own consensus by 9.5% on average, and by as much as 32% between individual pairs; on the same videos the pipeline differed from consensus by 7.5%. Automated counting is therefore already operating within the noise band of expert manual counting, and the practical question is not whether it is as accurate as a human counter but whether its errors are more predictable. The evidence here suggests they are: the counters vary in both directions with no group-level bias, whereas the pipeline overcounts consistently, and a consistent bias can be corrected while unstructured disagreement cannot. It also follows that a single manual count is a weak reference. ICC(2,1) of 0.816 for one counter against 0.947 for the mean of four means that programs validating automated tools against one person's counts risk attributing to the tool what is really counter noise.

A second contribution is the distance-based association fallback. Although the change is small, it had a large effect on footage recorded outside the format the detector was tuned for: together with a lower score threshold, it recovered 267 of the 459 urchins that the default configuration had missed on Big River Block-5 Heading-180. This suggests that when a detector is already available but footage differs from the original format, targeted improvements at the tracking stage can recover substantial performance far more cheaply than retraining the detector.

The most consequential practical finding is that capture consistency, rather than detector capacity, dominated accuracy. A change in resolution, aspect ratio, and frame rate alone was enough to drive a 62% undercount while the detector continued to find urchins perfectly well. This motivated the dive-camera field protocol in Appendix A. For restoration programs adopting automated counting, standardizing capture settings and diver movement is likely to yield a larger accuracy gain per unit of effort than any change to the model.

Several limitations remain. First, tracker settings were selected per video rather than fixed, so the reported errors describe tuned performance and bound from above what a single configuration would deliver; because the same seven videos were used to select those settings and to evaluate them, the errors are additionally optimistic by an unknown margin. Establishing one fixed profile per capture format, and validating it on videos not used for tuning, is the necessary next step. Second, only seven videos had manual reference counts suitable for quantitative validation, so the reported metrics should be treated as preliminary. Third, the detector was treated as a fixed component, so small, distant, and low-contrast urchins remain a likely source of error. Fourth, the density estimates depend on an unmeasured quantity: transect length is recorded, but the width of the strip a diver counts is not, and we bracket it at 1–2 m rather than measuring it. Because the transect tape is in frame by protocol and carries graduations, swath width could be measured directly from the footage.

5 Conclusion

We developed an automated pipeline for counting sea urchins from underwater transect videos, combining frame-level detection with Kalman filter-based tracking, Hungarian assignment, track confirmation, and a distance-based fallback for cases where IoU matching fails. Across the seven videos with manual reference counts the pipeline achieved a MAPE of 14.2%, against 9.5% disagreement among human counters on the videos all four scored, placing it inside the range of expert manual counting. On out-of-distribution footage the

distance fallback substantially reduced severe undercounting without requiring detector retraining.

The final output is not only a counting model but a reusable workflow: source code, parameter profiles, detection and pathway outputs, annotated review videos, and a diver filming protocol. Future work should fix one tracking profile per capture format and validate it on held-out videos, measure the counted swath directly from the transect tape, and fine-tune the detector on out-of-distribution footage, which we deliberately did not do here in order to isolate what the tracking stage alone can recover.

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A Dive-Camera Field Protocol

This protocol gives dive teams a one-page reference for camera settings, transect filming technique, filename conventions, and handling of non-standard formats. It was developed because the Big River results showed that inconsistent resolution, aspect ratio, frame rate, and camera motion introduce domain shift and increase the need for per-video parameter tuning.

Camera settings. Record at 1920×1080 , 16:9 aspect ratio (standard widescreen, not 8:7 or SuperView), 60 fps.

Field technique. Hover at the starting point for about 2 seconds with the camera steady and the substrate in view before beginning the swim. Swim forward only, along the transect tape, at a slow steady pace of roughly 0.25 m/s, keeping the camera at approximately constant altitude above the substrate and the tape visible in frame throughout for length verification. Do not stop and restart mid-transect, and do not backtrack; if you reverse and re-cross a region, urchins there will be counted twice. Hover at the endpoint for about 2 seconds before stopping the recording.

Filename convention. Rename offloaded files to the pattern

YYYY-MM-DD_<Site>_<Transect-or-Block>_<length>m.MP4

for example 2026-04-15_BigRiver_Block5_10m.MP4. The pipeline extracts the transect length directly from the filename; non-conforming names require manual entry.

Non-standard formats. Where a different format is unavoidable, for example a camera defaulting to 8:7, low light requiring 30 fps, or higher-resolution archival capture, the pipeline can still process the video but a format-specific parameter profile is required. Provide the resolution, aspect ratio, frame rate, a representative 30 s sample, and notes on unusual conditions.

Expected improvement. Surveys collected according to this protocol are expected to reduce boundary-effect undercount by approximately 5–10%, reduce between-survey variability introduced by inconsistent camera handling, and eliminate the need for per-video parameter tuning on in-distribution footage.